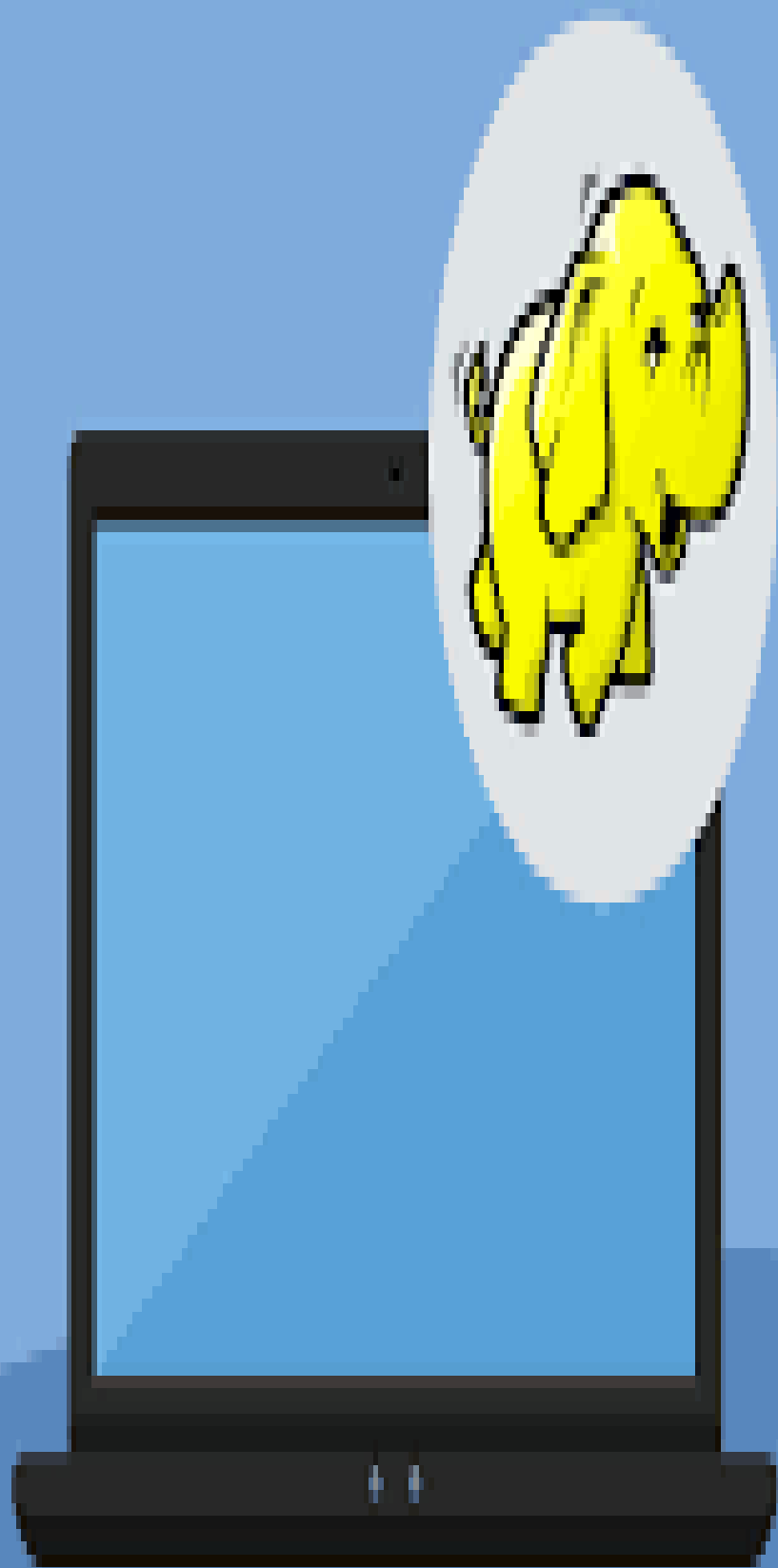




Hadoop Streaming

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INFORME PRÁCTICA DE HADOOP STREAMING

INTRODUCCIÓN

El presente trabajo tiene como objetivo principal la implementación y validación de un sistema de procesamiento de datos masivos en un entorno distribuido. Para ello, se utiliza el motor de Hadoop Streaming, una utilidad que permite la ejecución de trabajos de MapReduce utilizando scripts externos, en este caso desarrollados en el lenguaje de programación Python.

La práctica se estructura en dos fases diferenciadas que permiten observar el comportamiento de Hadoop ante distintos tipos de carga:

1. **Análisis de Texto Estructurado:** En primer lugar, se realiza un conteo de palabras (*WordCount*) sobre la obra completa de *El Quijote*, permitiendo verificar la correcta configuración del clúster y la lógica de los scripts mapper.py y reducer.py.
2. **Procesamiento de Datasets Complejos:** En la segunda fase, se procesa el dataset *20news-18828*, centrándose en categorías específicas como 'rec.autos', 'comp.sys.ibm.pc.hardware' y 'sci.space'. En esta etapa se introduce el concepto de **Stop Words** para filtrar términos gramaticales comunes y extraer las palabras con mayor relevancia temática de las noticias.

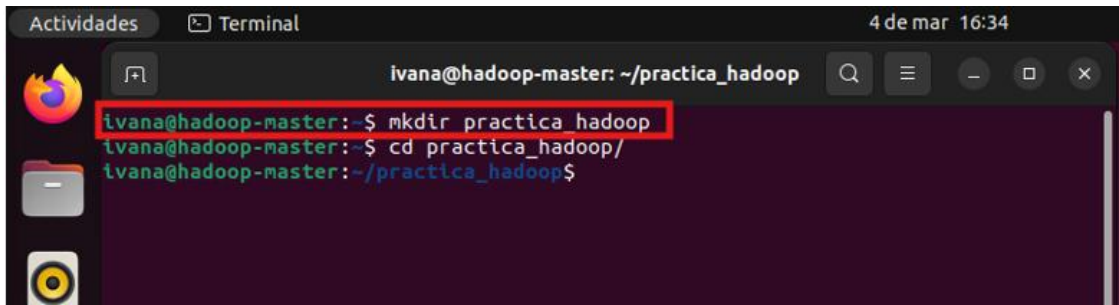
A través de esta experiencia práctica, se analizan no solo los resultados numéricos (rankings de frecuencia), sino también aspectos críticos del rendimiento del clúster, como el impacto del procesamiento de miles de archivos pequeños (*splits*) frente a la eficiencia esperada en sistemas de archivos distribuidos (HDFS).

PARTE 1: ANÁLISIS DE TEXTO DE EL QUIJOTE

Paso 1: Ubicación y creación de la carpeta de trabajo

Siguiendo la guía, se inició la terminal en el directorio `/home/ivana` y se creó una carpeta específica para organizar el proyecto.

- **Comando ejecutado:** `mkdir practica_hadoop && cd practica_hadoop`.
- **Ubicación:** Se confirmó que los archivos residen físicamente en el nodo `hadoop-master`.



A terminal window titled 'Terminal' with the date and time '4 de mar 16:34'. The prompt is 'ivana@hadoop-master: ~/practica_hadoop'. The user has entered three commands: 'mkdir practica_hadoop', 'cd practica_hadoop/', and the prompt has changed to 'ivana@hadoop-master: ~/practica_hadoop\$'. The first command is highlighted with a red box.

```
ivana@hadoop-master:~$ mkdir practica_hadoop
ivana@hadoop-master:~$ cd practica_hadoop/
ivana@hadoop-master:~/practica_hadoop$
```

Paso 2: Crear el script mapper.py

Se utilizó el editor de texto nano para implementar el estándar de Hadoop Streaming para el conteo de palabras, convirtiendo el texto a minúsculas y utilizando expresiones regulares para extraer palabras individuales.



A terminal window titled 'Terminal' with the date and time '8 de mar 10:58'. The prompt is 'ivana@hadoop-master: ~/practica_hadoop'. The user is editing a file named 'mapper.py' using the 'GNU nano 6.2' editor. The code in the file is as follows:

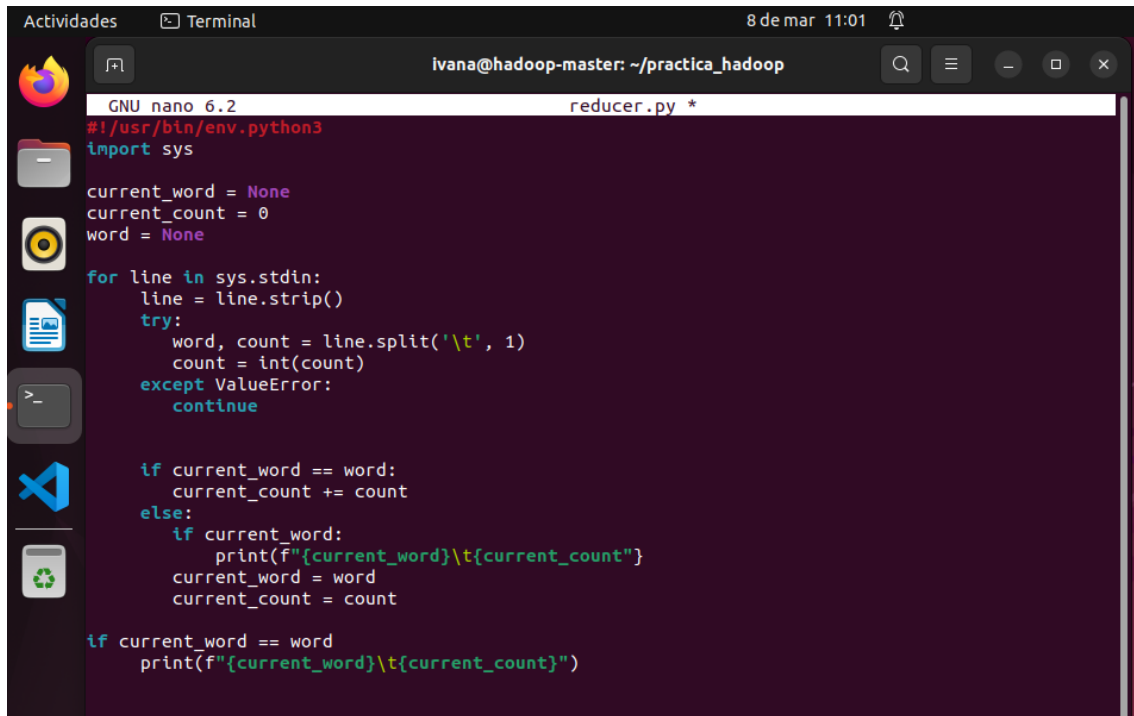
```
#!/usr/bin/env python3
import sys
import re

for line in sys.stdin:
    line = line.strip().lower()
    words = re.findall(r'[a-z0-9]+', line)
    for word in words:
        print(f"{word}\t1")
```

Paso 3: Crear el script reducer.py

Se procedió a crear el archivo encargado de la agregación de los datos.

- **Función:** El script lee la salida del mapper desde STDIN, suma las ocurrencias de cada palabra y emite el conteo final.



```
GNU nano 6.2 reducer.py *
#!/usr/bin/env python3
import sys

current_word = None
current_count = 0
word = None

for line in sys.stdin:
    line = line.strip()
    try:
        word, count = line.split('\t', 1)
        count = int(count)
    except ValueError:
        continue

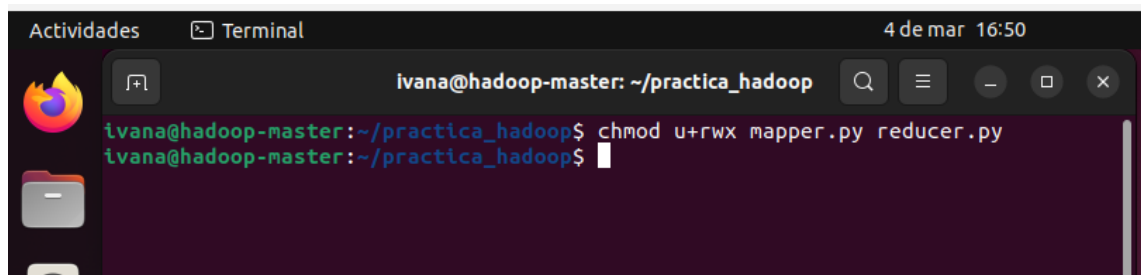
    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f"{current_word}\t{current_count}")
            current_word = word
            current_count = count

if current_word == word:
    print(f"{current_word}\t{current_count}")
```

Paso 4: Configurar permisos y preparar el entorno

Para que el motor de Hadoop pueda ejecutar los scripts de Python, se asignaron los permisos necesarios.

- **Comando:** `chmod u+rwx mapper.py reducer.py`.



```
ivana@hadoop-master: ~/practica_hadoop
ivana@hadoop-master:~/practica_hadoop$ chmod u+rwx mapper.py reducer.py
ivana@hadoop-master:~/practica_hadoop$
```

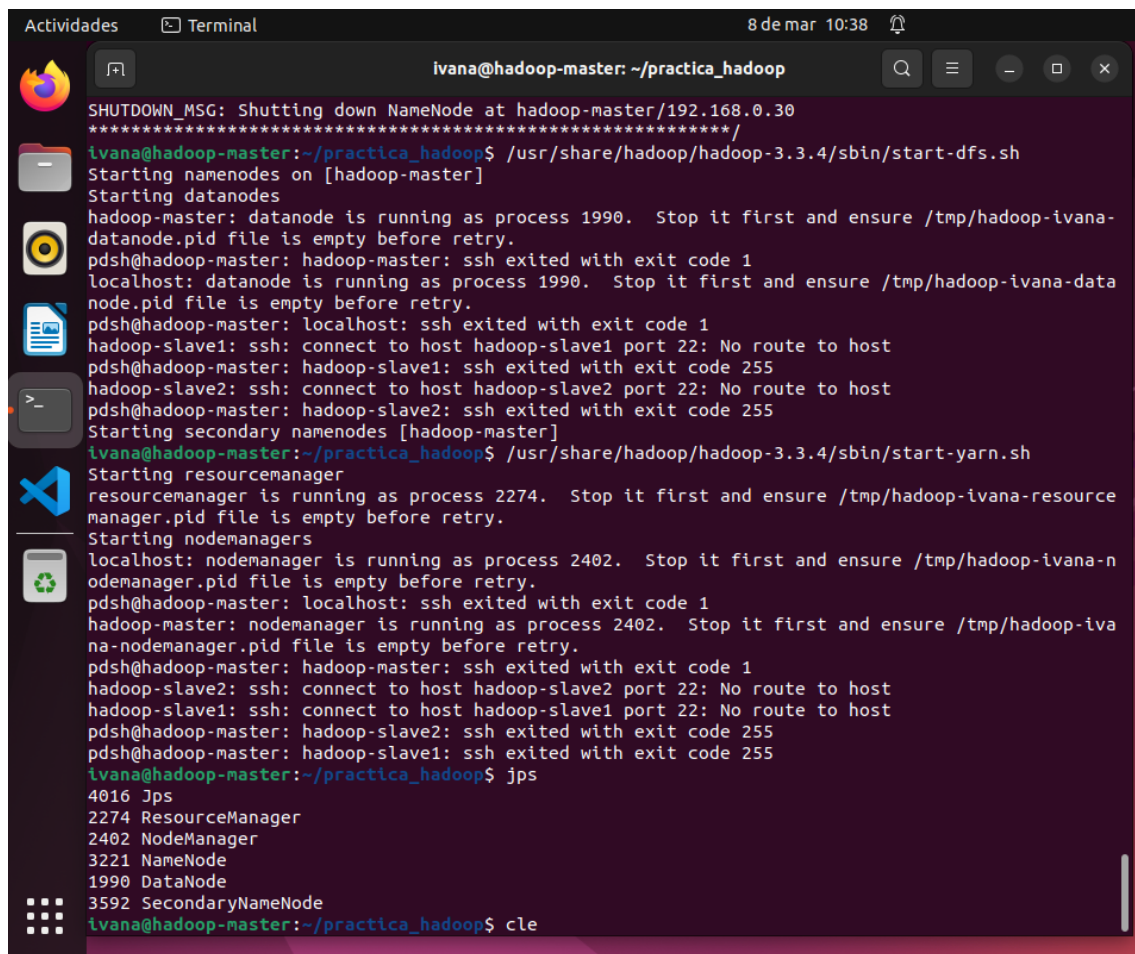
Paso 5: Ejecutar en el Clúster de Hadoop

Se levantaron los servicios distribuidos del clúster.

- **Arranque de servicios:** Se ejecutaron `start-dfs.sh` y `start-yarn.sh`.

```
/usr/share/hadoop/hadoop-3.3.4/sbin/start-dfs.sh
/usr/share/hadoop/hadoop-3.3.4/sbin/start-yarn.sh
```

- **Verificación:** Mediante el comando `jps` se verificó que el NameNode, DataNode, ResourceManager y NodeManager estuvieran activos.



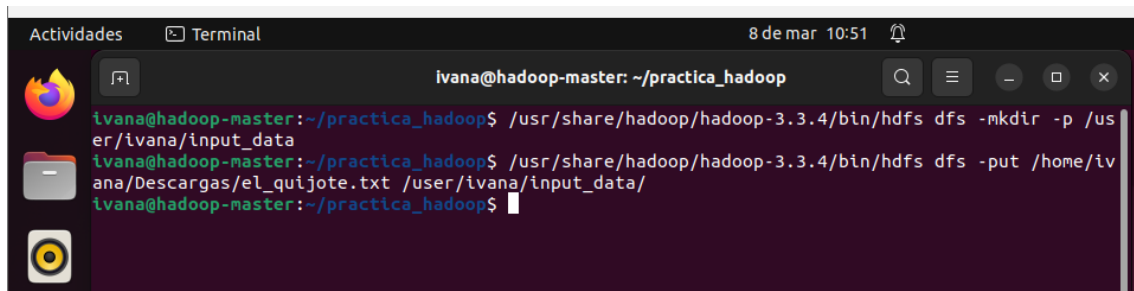
```
Actividades Terminal 8 de mar 10:38 ivana@hadoop-master: ~/practica_hadoop

SHUTDOWN_MSG: Shutting down NameNode at hadoop-master/192.168.0.30
*****
ivana@hadoop-master:~/practica_hadoop$ /usr/share/hadoop/hadoop-3.3.4/sbin/start-dfs.sh
Starting namenodes on [hadoop-master]
Starting datanodes
hadoop-master: datanode is running as process 1990. Stop it first and ensure /tmp/hadoop-ivana-
datanode.pid file is empty before retry.
pdsh@hadoop-master: hadoop-master: ssh exited with exit code 1
localhost: datanode is running as process 1990. Stop it first and ensure /tmp/hadoop-ivana-data
node.pid file is empty before retry.
pdsh@hadoop-master: localhost: ssh exited with exit code 1
hadoop-slave1: ssh: connect to host hadoop-slave1 port 22: No route to host
pdsh@hadoop-master: hadoop-slave1: ssh exited with exit code 255
hadoop-slave2: ssh: connect to host hadoop-slave2 port 22: No route to host
pdsh@hadoop-master: hadoop-slave2: ssh exited with exit code 255
Starting secondary namenodes [hadoop-master]
ivana@hadoop-master:~/practica_hadoop$ /usr/share/hadoop/hadoop-3.3.4/sbin/start-yarn.sh
Starting resource manager
resource manager is running as process 2274. Stop it first and ensure /tmp/hadoop-ivana-resource
manager.pid file is empty before retry.
Starting node managers
localhost: nodemanager is running as process 2402. Stop it first and ensure /tmp/hadoop-ivana-n
odemanager.pid file is empty before retry.
pdsh@hadoop-master: localhost: ssh exited with exit code 1
hadoop-master: nodemanager is running as process 2402. Stop it first and ensure /tmp/hadoop-iva
na-nodemanager.pid file is empty before retry.
pdsh@hadoop-master: hadoop-master: ssh exited with exit code 1
hadoop-slave2: ssh: connect to host hadoop-slave2 port 22: No route to host
hadoop-slave1: ssh: connect to host hadoop-slave1 port 22: No route to host
pdsh@hadoop-master: hadoop-slave2: ssh exited with exit code 255
pdsh@hadoop-master: hadoop-slave1: ssh exited with exit code 255
ivana@hadoop-master:~/practica_hadoop$ jps
4016 Jps
2274 ResourceManager
2402 NodeManager
3221 NameNode
1990 DataNode
3592 SecondaryNameNode
ivana@hadoop-master:~/practica_hadoop$ cle
```

Paso 6: Carga de datos y ejecución del Job

Se procedió a subir el archivo de *El Quijote* al sistema de archivos distribuido (HDFS).

- **Comandos HDFS:**
 1. **Crear carpeta:** `/usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -mkdir -p /user/ivana/input_data`
 2. **Subir archivo:** `/usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -put /home/ivana/Descargas/el_quijote.txt /user/ivana/input_data/`
- **Lanzamiento del Job:** Se ejecutó el motor de streaming apuntando al JAR `hadoop-streaming-3.3.4.jar` y definiendo los archivos de mapper y reducer



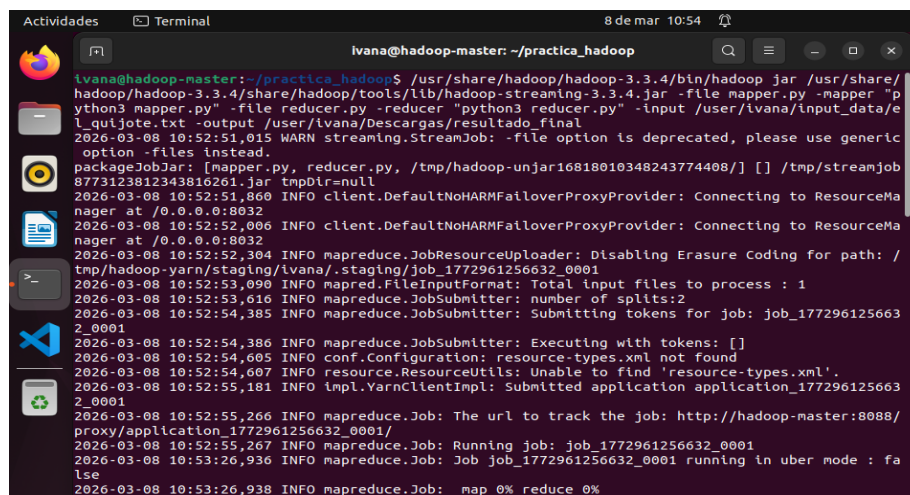
```

ivana@hadoop-master: ~/practica_hadoop
ivana@hadoop-master:~/practica_hadoop$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -mkdir -p /user/ivana/input_data
ivana@hadoop-master:~/practica_hadoop$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -put /home/ivana/Descargas/el_quijote.txt /user/ivana/input_data/
ivana@hadoop-master:~/practica_hadoop$
  
```

Paso 7: Resultados y Ranking

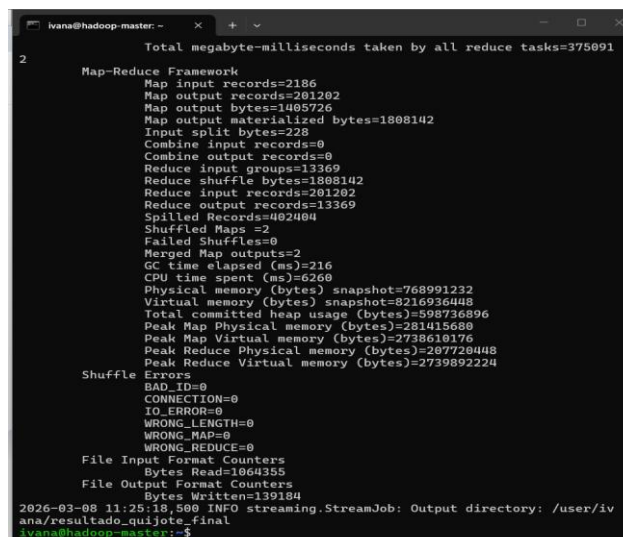
1. Lanzamos el comando final:

`/usr/share/hadoop/hadoop-3.3.4/bin/hadoop jar /usr/share/hadoop/hadoop-3.3.4/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -file mapper.py -mapper "python3 mapper.py" -file reducer.py -reducer "python3 reducer.py" -input /user/ivana/input_data/el_quijote.txt -output /user/ivana/resultado_final`



```

ivana@hadoop-master: ~/practica_hadoop
ivana@hadoop-master:~/practica_hadoop$ /usr/share/hadoop/hadoop-3.3.4/bin/hadoop jar /usr/share/hadoop/hadoop-3.3.4/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -file mapper.py -mapper "python3 mapper.py" -file reducer.py -reducer "python3 reducer.py" -input /user/ivana/input_data/el_quijote.txt -output /user/ivana/resultado_final
2026-03-08 10:52:51,015 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper.py, reducer.py, /tmp/hadoop-unjar1681801034824377408/] [] /tmp/streamjob8773123812343816261.jar tmpDir=null
2026-03-08 10:52:51,860 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-08 10:52:52,006 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-08 10:52:52,304 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ivana/.staging/job_1772961256632_0001
2026-03-08 10:52:53,090 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-08 10:52:53,616 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-08 10:52:54,385 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1772961256632_0001
2026-03-08 10:52:54,386 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-08 10:52:54,605 INFO conf.Configuration: resource-types.xml not found
2026-03-08 10:52:54,607 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-08 10:52:55,181 INFO impl.YarnClientImpl: Submitted application application_1772961256632_0001
2026-03-08 10:52:55,266 INFO mapreduce.Job: The url to track the job: http://hadoop-master:8088/proxy/application_1772961256632_0001/
2026-03-08 10:52:55,267 INFO mapreduce.Job: Running job: job_1772961256632_0001
2026-03-08 10:53:26,936 INFO mapreduce.Job: Job job_1772961256632_0001 running in uber mode : false
2026-03-08 10:53:26,938 INFO mapreduce.Job: map 0% reduce 0%
  
```



```

ivana@hadoop-master: ~/practica_hadoop
ivana@hadoop-master:~/practica_hadoop$ /usr/share/hadoop/hadoop-3.3.4/bin/hadoop jar /usr/share/hadoop/hadoop-3.3.4/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -file mapper.py -mapper "python3 mapper.py" -file reducer.py -reducer "python3 reducer.py" -input /user/ivana/input_data/el_quijote.txt -output /user/ivana/resultado_final
2026-03-08 11:25:18,500 INFO streaming.StreamJob: Output directory: /user/ivana/resultado_final
ivana@hadoop-master:~/practica_hadoop$
  
```

```

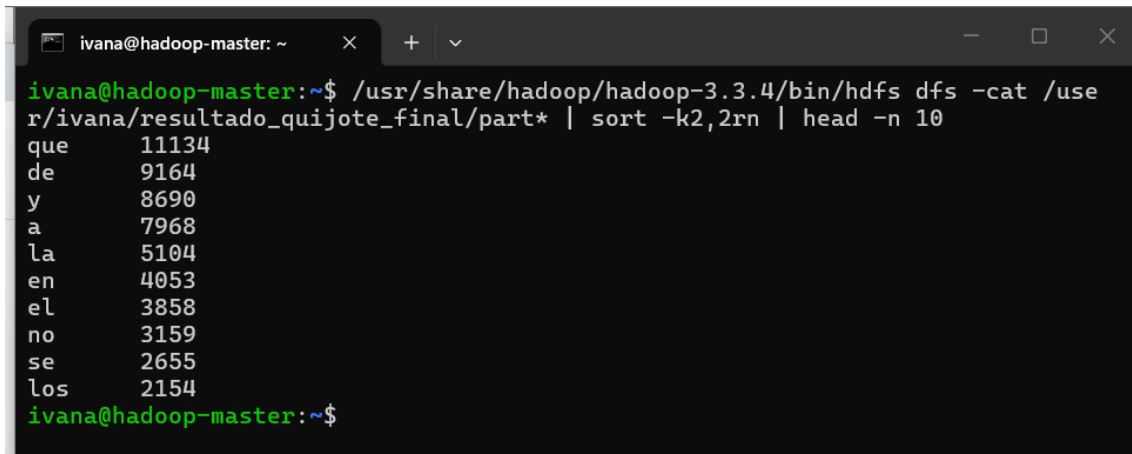
Total megabyte-milliseconds taken by all reduce tasks=375091
2
Map-Reduce Framework
  Map input records=2186
  Map output records=201202
  Map output bytes=1405726
  Map output materialized bytes=1808142
  Input split bytes=228
  Combine input records=0
  Combine output records=0
  Reduce input groups=13369
  Reduce shuffle bytes=1808142
  Reduce input records=201202
  Reduce output records=13369
  Spilled Records=102404
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=216
  CPU time spent (ms)=6260
  Physical memory (bytes) snapshot=768991232
  Virtual memory (bytes) snapshot=8216936448
  Total committed heap usage (bytes)=598736896
  Peak Map Physical memory (bytes)=291415600
  Peak Map Virtual memory (bytes)=2738610176
  Peak Reduce Physical memory (bytes)=207728448
  Peak Reduce Virtual memory (bytes)=2739892224
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=1064355
File Output Format Counters
  Bytes Written=139184
2026-03-08 11:25:18,500 INFO streaming.StreamJob: Output directory: /user/ivana/resultado_final
ivana@hadoop-master:~/practica_hadoop$
  
```

2- Ver el Ranking

Una vez finalizado el trabajo, procedemos a extraer el ranking solicitado en la práctica.

- **Comando de visualización:**

hdfs dfs -cat /user/ivana/resultado_quijote_final/part | sort -k2,2rn | head -n 10.*



```
ivana@hadoop-master: ~  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -cat /user/ivana/resultado_quijote_final/part* | sort -k2,2rn | head -n 10  
que      11134  
de        9164  
y         8690  
a         7968  
la        5104  
en        4053  
el        3858  
no        3159  
se        2655  
los       2154  
ivana@hadoop-master:~$
```

Ranking Top 10 de Palabras (El Quijote)

Palabra	Conteo Total
que	11.134
de	9.164
y	8.690
a	7.968
la	5.104

Palabra	Conteo Total
en	4.053
el	3.858
no	3.159
se	2.655
los	2.154

Nota final: Para concluir la entrega, se ha descargado el resultado a la carpeta local de Linux para su envío final.

```
ivana@hadoop-master: ~  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -get /user/ivana/resultado_quijote_final/part-00000 ~/practica_hadoop/resultado_quijote.txt  
ivana@hadoop-master:~$ |
```

2: PROCESAMIENTO DEL DATASET 20NEWS-18828

Esta parte de la tarea consistió en aplicar el mismo proceso a un dataset de noticias mucho más complejo, filtrando categorías específicas y eliminando palabras vacías (*stop words*)

Preparación del Dataset

- Se descomprimió el archivo 20news-18828.tar.gz.

- Se seleccionaron y subieron a HDFS las categorías indicadas: rec.autos, comp.sys.ibm.pc.hardware y sci.space.

```
ivana@hadoop-master: ~  
ivana@hadoop-master:~$ tar -xzf ~/Descargas/20news-18828.tar.gz -C ~/practica_hadoop  
ivana@hadoop-master:~$
```

```
ivana@hadoop-master: ~  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -rm -r /user/ivana/input_news  
Deleted /user/ivana/input_news  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -mkdir -p /user/ivana/input_news  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -put ~/practica_hadoop/20news-18828/rec.autos/* /user/ivana/input_news/  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -put ~/practica_hadoop/20news-18828/comp.sys.ibm.pc.hardware/* /user/ivana/input_news/  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -put ~/practica_hadoop/20news-18828/sci.space/* /user/ivana/input_news/
```

Modificación del Mapper para filtrar "Stop Words"

Para que el ranking tenga sentido (y no salgan palabras como "que" o "de"), vamos a incluir una lista de palabras a ignorar directamente en el mapper.py.

```
GNU nano 6.2 mapper.py *  
#!/usr/bin/env python3  
import sys  
import re  
  
# Lista manual de stop words comunes  
STOP_WORDS = {  
    'que', 'de', 'y', 'a', 'la', 'en', 'el', 'no', 'se', 'los',  
    'las', 'un', 'por', 'con', 'para', 'su', 'al', 'lo', 'como',  
    'the', 'to', 'and', 'of', 'a', 'in', 'is', 'it', 'for'  
}  
  
for line in sys.stdin:  
    line = line.strip().lower()  
    # Extraer palabras (letras y números)  
    words = re.findall(r'[a-z0-9]+', line)  
    for word in words:  
        # Solo enviamos la palabra si no es una stop word y tiene más de 2  
        if word not in STOP_WORDS and len(word) > 2:  
            print(f"{word}\t1")
```

Ejecución y Rendimiento

Lanzamos el comando de Hadoop Streaming apuntando a la nueva carpeta de entrada.

```

ivana@hadoop-master: ~$ /usr/share/hadoop/hadoop-3.3.4/bin/hadoop jar /usr/share/hadoop/hadoop-3.3.4/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -file mapper.py -mapper "python3 mapper.py" -file reducer.py -reducer "python3 reducer.py" -input /user/ivana/input_news -output /user/ivana/resultado_noticias_top

```

```

ivana@hadoop-master: ~$
2026-03-08 12:07:09,124 INFO mapreduce.Job: The url to track the job: http://hadoop-master:8088/proxy/application_1772961256632_0006/
2026-03-08 12:07:09,128 INFO mapreduce.Job: Running job: job_1772961256632_0006
2026-03-08 12:07:37,573 INFO mapreduce.Job: Job job_1772961256632_0006 running in uber mode : false
2026-03-08 12:07:37,575 INFO mapreduce.Job: map 0% reduce 0%
2026-03-08 12:08:31,398 INFO mapreduce.Job: map 1% reduce 0%
2026-03-08 12:09:40,952 INFO mapreduce.Job: map 2% reduce 0%
2026-03-08 12:10:55,425 INFO mapreduce.Job: map 3% reduce 0%
2026-03-08 12:12:06,461 INFO mapreduce.Job: map 4% reduce 0%
2026-03-08 12:13:19,044 INFO mapreduce.Job: map 5% reduce 0%
2026-03-08 12:14:31,823 INFO mapreduce.Job: map 6% reduce 0%
2026-03-08 12:15:53,051 INFO mapreduce.Job: map 7% reduce 0%
2026-03-08 12:17:00,261 INFO mapreduce.Job: map 8% reduce 0%
2026-03-08 12:18:12,651 INFO mapreduce.Job: map 9% reduce 0%
2026-03-08 12:19:21,473 INFO mapreduce.Job: map 10% reduce 0%
2026-03-08 12:20:35,450 INFO mapreduce.Job: map 11% reduce 0%
2026-03-08 12:21:46,268 INFO mapreduce.Job: map 12% reduce 0%
2026-03-08 12:22:56,987 INFO mapreduce.Job: map 13% reduce 0%
2026-03-08 12:24:17,235 INFO mapreduce.Job: map 14% reduce 0%
2026-03-08 12:25:24,992 INFO mapreduce.Job: map 15% reduce 0%
2026-03-08 12:26:34,741 INFO mapreduce.Job: map 16% reduce 0%
2026-03-08 12:27:44,273 INFO mapreduce.Job: map 17% reduce 0%
2026-03-08 12:28:27,053 INFO mapreduce.Job: map 17% reduce 6%
2026-03-08 12:28:57,687 INFO mapreduce.Job: map 18% reduce 6%
2026-03-08 12:30:12,164 INFO mapreduce.Job: map 19% reduce 6%
2026-03-08 12:31:28,562 INFO mapreduce.Job: map 20% reduce 6%
2026-03-08 12:31:32,925 INFO mapreduce.Job: map 20% reduce 7%
2026-03-08 12:32:42,996 INFO mapreduce.Job: map 21% reduce 7%
2026-03-08 12:33:58,219 INFO mapreduce.Job: map 22% reduce 7%
2026-03-08 12:35:07,603 INFO mapreduce.Job: map 23% reduce 7%
2026-03-08 12:35:08,619 INFO mapreduce.Job: map 23% reduce 8%
2026-03-08 12:36:19,531 INFO mapreduce.Job: map 24% reduce 8%
2026-03-08 12:37:34,281 INFO mapreduce.Job: map 25% reduce 8%
2026-03-08 12:38:35,958 INFO mapreduce.Job: map 26% reduce 8%
2026-03-08 12:38:42,244 INFO mapreduce.Job: map 26% reduce 9%
2026-03-08 12:39:49,968 INFO mapreduce.Job: map 27% reduce 9%
2026-03-08 12:41:08,941 INFO mapreduce.Job: map 28% reduce 9%
2026-03-08 12:42:19,327 INFO mapreduce.Job: map 29% reduce 9%

```

```

ivana@hadoop-master: ~$
7952
Total megabyte-milliseconds taken by all reduce tasks=613183
Map-Reduce Framework
  Map input records=83018
  Map output records=275524
  Map output bytes=2542007
  Map output materialized bytes=3107827
  Input split bytes=259500
  Combine input records=0
  Combine output records=0
  Reduce input groups=23204
  Reduce shuffle bytes=3107827
  Reduce input records=275524
  Reduce output records=23204
  Spilled Records=551048
  Shuffled Maps =2462
  Failed Shuffles=0
  Merged Map outputs=2462
  GC time elapsed (ms)=658984
  CPU time spent (ms)=1890420
  Physical memory (bytes) snapshot=70205335520
  Virtual memory (bytes) snapshot=6730149986304
  Total committed heap usage (bytes)=626771623936
  Peak Map Physical memory (bytes)=340942848
  Peak Map Virtual memory (bytes)=2753011712
  Peak Reduce Physical memory (bytes)=232738816
  Peak Reduce Virtual memory (bytes)=2756186112
Shuffle
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=3532262
File Output Format Counters
  Bytes Written=243246
2026-03-08 14:07:53,936 INFO streaming.StreamJob: Output directory: /user/ivana/resultado_noticias_top
ivana@hadoop-master: ~$

```

Generación del Ranking Final

```
ivana@hadoop-master: ~  
ivana@hadoop-master:~$ /usr/share/hadoop/hadoop-3.3.4/bin/hdfs dfs -get /user/ivana/resultado_noticias_top/part-00000 ~/practica_hadoop/resultado_final_tarea.txt  
ivana@hadoop-master:~$  
  
ivana@hadoop-master:~$ ls /home/ivana/practica_hadoop/  
20news-18828 reducer.py resultado_quijote.txt  
mapper.py resultado_final_tarea.txt  
ivana@hadoop-master:~$ |
```

Ranking de las Categorías de Noticias

Tras completar el Job application_1772961256632_0006, se generó el ranking de las 10 palabras más frecuentes en los grupos de noticias seleccionados.

Posición	Palabra	Frecuencia
1	from	2.531
2	subject	2.479
3	lines	2.464
4	organization	2.378
5	writes	1.849
6	article	1.706

Posición	Palabra	Frecuencia
7	space	1.131
8	university	1.054
9	nntp	986
10	posting	983

Conclusión

Se ha cumplido con el objetivo de la práctica al procesar un dataset complejo de forma distribuida. La diferencia de tiempos entre la Parte 1 y la Parte 2 evidencia el problema de los "archivos pequeños" en Hadoop, mientras que el ranking de noticias demuestra la efectividad del filtrado de palabras para extraer términos temáticos como "space" u "organization".